Experiment 1: Flashing of LED using AT89C51 using Proteus

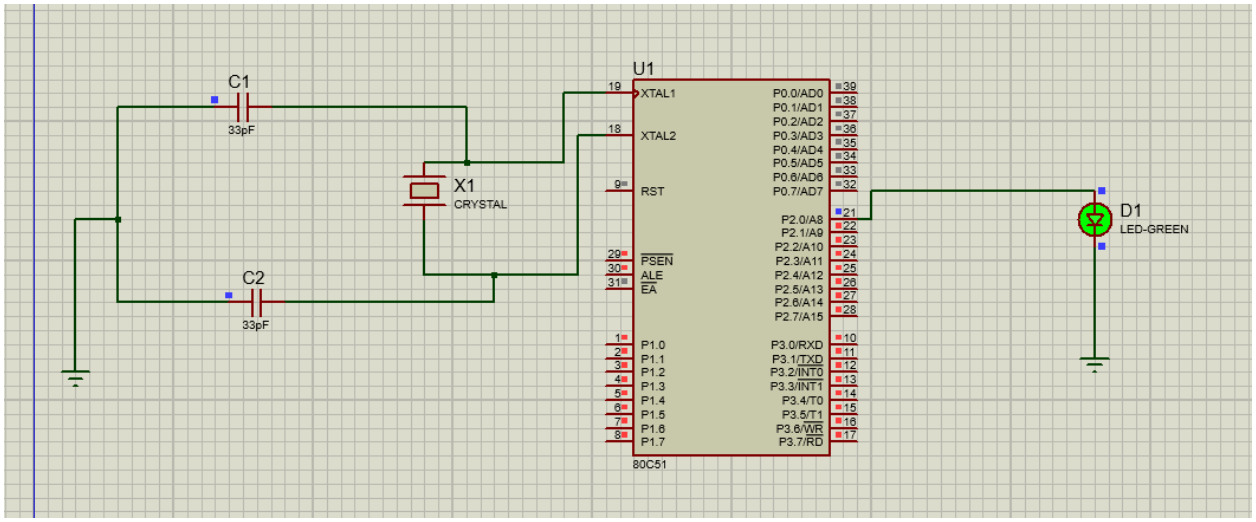
Aim:

To design and simulate LED flashing using the AT89C51 microcontroller in Proteus.

Procedure:

1. Open Proteus and create a new circuit.
2. Place AT89C51, LED, resistor and power supply.
3. Connect the LED to a suitable port of AT89C51.
4. Write the LED blinking program and compile it.
5. Load the HEX file into AT89C51.
6. Run the simulation and observe the LED.

```
1  ORG 0000H
2
3  UP:     SETB P2.0       ; Turn ON LED
4          ACALL DELAY
5          CLR P2.0        ; Turn OFF LED
6          ACALL DELAY
7          SJMP UP         ; Repeat continuously
8
9  DELAY:  MOV R4,#35
10 H1:     MOV R3,#255
11 H2:     DJNZ R3,H2
12         DJNZ R4,H1
13         RET
14
15 END
```



Result:

The LED was successfully flashed at regular intervals using AT89C51 in Proteus.